that involved introducing an arbitrary cutoff on the energies of the field vibrations we were considering. The Bekenstein-Hawking entropy, on the other hand, is just a finite number, full stop. For another thing, the entanglement entropy in field theory should depend on exactly how many fields are involved—the electrons, quarks, neutrinos, and so forth. The formula for black-hole entropy that Hawking derived makes no mention of such things at all.

If we can't simply attribute black-hole entropy to the quantum fields inside, the alternative is to imagine that spacetime itself is made of some quantum degrees of freedom, and the Bekenstein-Hawking formula measures the entanglement of the degrees of freedom inside the black hole with the degrees of freedom outside. If that sounds pretty vague, that's because it is. We're not precisely sure what these spacetime degrees of freedom are, or how they interact with one another. But the general principles of quantum mechanics should still be respected. If there's entropy, and that entropy comes from entanglement, there must be degrees of freedom that can entangle with the rest of the world in many different ways, even if classical black holes are all featureless.

If this story is right, the number of degrees of freedom in a black hole isn't infinite, but it is very large indeed. Our Milky Way galaxy contains a supermassive black hole at its center, associated with a radio source called Sagittarius A*. From observing how stars orbit around the hole, we can measure its mass to be 4 million times the mass of the sun. That corresponds to an entropy of 10^{90}, which is greater than the entropy of all the known particles in the entire observable universe. The number of degrees of freedom in a quantum system has to be at least as large as its entropy, since that entropy comes precisely from those degrees of freedom being entangled with the outside world. So there must be at least 10^{90} degrees of freedom in the black hole.

While we tend to pay attention to the stuff we see in the universe—matter, radiation, and so on—almost all of the universe's quantum degrees of freedom are invisible, doing nothing more than stitching spacetime together. In a volume of space roughly the size of an adult human, there must be at least 10^{70} degrees of freedom; we know that because that's the entropy of a black hole that would fill such a volume. But there are only about 10^{28} particles in a person. We can think of a particle as a degree of freedom that has been "turned on," while all the other degrees of freedom